

Guneev Pannu

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EDUCATION

University of Toronto – 3.96 GPA

Sept 2024 – May 2028

HBSc Double Major in Computer Science and Statistics, Math Minor

Relevant Courses: Data Structures & Algorithms, Object Oriented Development, Systems Programming, Databases

EXPERIENCE

Teaching Assistant

Sept 2026 – Dec 2026

University of Toronto

- Will lead weekly tutorials for **20+** students, teaching foundational programming concepts in Python for CSC108

Software Engineer

May 2026 – Aug 2026

University of Toronto

- Architected a live queuing and logging system for office hours, securing a Fall rollout with **500+ students**
- Automated student enrollment, saving **2+ hrs** of manual entry for professors by building a CSV pipeline
- Developed a Cron based email reminder service for interested students, increasing office hour attendance by **15%**
- Integrated UofT SAML authentication for **15K+ students**, eliminating manual account creation

Computer Science Education Research Assistant

May 2026 – June 2026

University of Toronto

- Authored weekly lab materials and slide decks in LaTeX to deepen understanding of **500+ students** for CSC108
- Streamlined Python homework by cutting **33%** of existing redundant questions and replacing with **38** new questions
- Automated code evaluation achieving **100%** code coverage with pytest test suite for class assignments

Web Developer

Sept 2025 – May 2026

UofT Machine Learning Club

- Shipped 11 web pages to publicize the talent of **400+ club members** in Next.js
- Developed a content management system through custom forms to allow editing of pages by non-tech people
- Developed an interview dashboard system to streamline recruitment for **40+ applicants**

PROJECTS

Blog Website Backend | *Spring Boot, Java, Maven, Git, PostgreSQL, Docker*

- Built a backend software application in a week that could support a blog post social media app, like Tumblr
- Integrated Spring Data JPA to simplify communication from the service layer to the persistence layer
- Implemented JWT-based authentication and role-based authorization using Spring Security
- Integrated development tools like Lombok and MapStruct to reduce code boiler plater speeding development.

Paint Program | *Java, JavaFX, Maven, Git*

- Built a paint program that allowed users to draw **180+ shapes** and lines on a resizable canvas using **JavaFX**
- Applied the **MVC** design pattern reducing merge conflicts by **30%**, allowing for more rapid development
- Engineered a save-load system of the current canvas using finite state machines to achieve load times of **<100 ms**

Custom Unix Shell | *C, Make, Git*

- Developed a custom **Unix shell** in **C** replicating core Bash functionality, including process creation, piping, and command parsing
- Implemented inter-process communication and socket-based client-server chat system over local networks
- Debugged and memory-optimized system using **GDB** and **Valgrind**, achieving zero memory leaks

TECHNICAL SKILLS

Languages: Java, Python, C, Typescript

Frameworks: React, Node.js, Junit, Springboot, Agile, Next.js

Developer Tools: Git, Docker, PostgrsSQL, Postman, Linux, Cursor

Libraries: pytest, Prisma, PyTorch, JavaFX